

AJAY H

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CAREER OBJECTIVE

B.Tech in Electronics and Communication Engineering with specialized training in Embedded Systems at Emertxe Information Technologies, Bangalore. Proficient in C programming, Linux OS, and Linux Internals, with hands-on experience in embedded system development. Seeking to contribute to real-world embedded solutions in a forward-thinking electronics environment.

WORK EXPERIENCE

- Currently undergoing technical training program – **Advanced Embedded Systems Course** at Emertxe Information Technologies (<http://www.emertxe.com>) Bangalore.

TECHNICAL SKILLS

- Programming & Concepts** : C Programming • Data Structures & Algorithms • Linux Internals • Embedded C • Microcontroller Programming
- System Programming** : Linux Kernel System Calls • IPC (Pipes, FIFO, Shared Memory) • TCP/UDP Socket Programming • Multithreading
- Embedded Platforms** : PIC18F4580 • ATmega328P • Raspberry Pi
- Protocols** : UART • SPI • I2C • CAN
- Tools & IDEs** : MPLAB X IDE • GCC • Makefile • VSCode

EDUCATION

- B.Tech (ECE) | APJ Abdul Kalam Technological University (KTU) | 76.10% | 2025
- Class – XII | CBSE | 79% | 2021
- Class – X | CBSE | 93.40% | 2019

CONTRIBUTIONS AND ACHIEVEMENTS

- My academic Project “TALK TO WRITE ASSIST” received national and international media coverage, featured on NDTV, The Indian Express, Brut India, and other digital platforms.
 - [[NDTV](#)] | [[The Indian Express](#)] | [[Brut India](#)]
- Participated in Project Expo at “ENTE KERALAM 2025”, conducted by the Kerala Government at Alappuzha, showcasing innovative technical project.

PROJECTS

1 .Title

CAN-Based Automotive Dashboard System

Project brief

Developed a CAN bus prototype with three ECUs — RPM/indicator, speed/gear, and a display node — simulating real-time automotive dashboard communication with accurate frame formatting and message parsing.

2 .Title

TFTP – Trivial File Transfer Protocol (Client–Server Implementation)

Project brief

Designed and implemented a TFTP-based client-server application enabling file transfer over UDP, supporting file upload (WRQ) and download (RRQ) operations.

Focused on protocol-compliant packet formatting, acknowledgment-based reliable transfer, and robust error handling.

3 .Title

MiniShell – Custom Linux Command Line Interpreter

Project brief

Built a Linux-based MiniShell capable of parsing and executing internal/external commands with argument support, process creation, and termination using system calls.

4 .Title

Inverted Search

Project brief

Built a C-based Inverted Search System using hash tables and linked lists to index and search words across multiple files, returning match locations and occurrence counts.

5 .Title

Arbitrary Precision Calculator

Project brief

Built a C-based APC using doubly linked lists to handle unlimited-precision integers, supporting all four arithmetic operations with positive and negative number handling.

6 .Title

LSB Image Steganography - Encoding and Decoding

Project brief

Built a C-based LSB steganography tool to embed and extract secret text within BMP images, encoding data length and payload into image bytes for secure covert transmission.

7 .Title

Mp3 Tag Reader and Editor

Project brief

Developed a command-line tool in C to read and modify ID3 metadata tags in MP3 files — including title, artist, album, and genre — ensuring audio content remains intact during updates.

8 .Title

Address Book Management System

Project brief

Developed a menu-driven Address Book application in C for efficient contact management, supporting operations including add, search, edit, delete, and display of contact records.

ACADEMIC PROJECTS

1 .Title

Talk-to-Write Assist

Project brief

Built a speech-to-handwriting assistive system integrating AI speech recognition with CNC motion control, converting spoken words into G-code for physical text output via a pen plotter.

2 .Title

Anti-Collision Real timeWarning System with Hybrid Power Generation

Project brief

Developed a road safety system for hairpin bends that detects and classifies approaching vehicles using load cells, triggering real-time LED-based alerts for oncoming drivers. Powered by a hybrid energy system combining solar panels and piezoelectric sensors for uninterrupted operation in remote environments.